

SESSION-11:

Tasks

1.

Declare an integer variable called likes and a pointer variable called ptrLikes; assign likes a value, point ptrLikes to likes, and print both the value and the address stored in ptrLikes.

```
#include <stdio.h>
```

```
int main() {
```

```
    int likes = 500;
```

```
    int *ptrLikes;
```

```
    ptrLikes = &likes;
```

```
    printf("Likes: %d\n", likes);
```

```
    printf("Address stored in ptrLikes: %p\n", (void *)ptrLikes);
```

```
    return 0;
```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> gcc task1_pointer.c -o task1_pointer
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> .\task1_pointer.exe
Likes: 500
Address stored in ptrLikes: 000000BDFE5FF744
```

2.

Write a function swapPlaylistCounts(int *a, int *b) that swaps the number of songs in two Spotify playlists using pointers, then call the function in main and print the swapped values.

```
#include <stdio.h>
```

```
void swapPlaylistCounts(int *a, int *b) {
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```

int main() {
    int playlist1 = 25;
    int playlist2 = 40;
    printf("Before swapping:\n");
    printf("Playlist 1: %d songs\n", playlist1);
    printf("Playlist 2: %d songs\n", playlist2);
    swapPlaylistCounts(&playlist1, &playlist2);
    printf("\nAfter swapping:\n");
    printf("Playlist 1: %d songs\n", playlist1);
    printf("Playlist 2: %d songs\n", playlist2);
    return 0;
}

```

```

PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> gcc task2_swap.c -o task2_swap
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> .\task2_swap.exe
Before swapping:
Playlist 1: 25 songs
Playlist 2: 40 songs

After swapping:
Playlist 1: 40 songs
Playlist 2: 25 songs

```

3.

Given an array of 5 order amounts (e.g., Zomato orders), use a pointer to iterate through the array and print each amount along with its memory address.

Hint: Use pointer arithmetic to move to the next element.

```
#include <stdio.h>
```

```

int main() {
    int orders[5] = {250, 450, 300, 600, 150};
    int *ptr = orders;
    for (int i = 0; i < 5; i++) {

```

```

    printf("Order amount: %d, Address: %p\n",
           *(ptr + i), (void *) (ptr + i));
}

return 0;
}

```

```

PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> gcc task3_array_pointer.c -o task3_array_pointer
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> .\task3_array_pointer.exe
Order amount: 250, Address: 000000A107DFFBD0
Order amount: 450, Address: 000000A107DFFBD4
Order amount: 300, Address: 000000A107DFFBD8
Order amount: 600, Address: 000000A107DFFBDC
Order amount: 150, Address: 000000A107DFFBE0

```

4.

Create a function `incrementFollowers(int *followers, int n)` that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.

```
#include <stdio.h>
```

```

void incrementFollowers(int *followers, int n) {
    for (int i = 0; i < n; i++) {
        *(followers + i) += 100;
    }
}

```

```

int main() {
    int followers[5] = {1200, 2500, 3400, 1800, 5000};
    incrementFollowers(followers, 5)
    printf("Updated follower counts:\n");
    for (int i = 0; i < 5; i++) {
        printf("Friend %d: %d followers\n", i + 1, followers[i]);
    }
    return 0;
}

```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> gcc task4_followers.c -o task4_followers
PS C:\Users\SIDDHI\OneDrive\Desktop\C_Assignment> .\task4_followers.exe
Updated follower counts:
Friend 1: 1300 followers
Friend 2: 2600 followers
Friend 3: 3500 followers
Friend 4: 1900 followers
Friend 5: 5100 followers
```

5.

Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example; label the variable, pointer, and address.

